

**New trading  
relationships can  
change the  
dominant vehicle.**



**Centre for Blockchain  
Technologies**

# **The Making of Dominant Vehicle Currencies**

Evidence from DeFi  
Jiahua Xu

# A cross-border payment needs someone to make the FX market

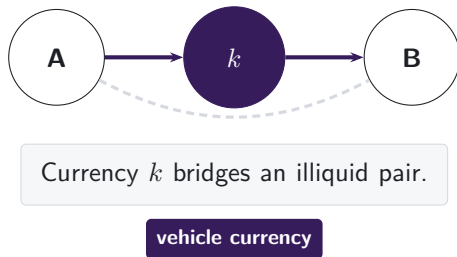
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## THE LIQUIDITY-PROVIDER PROBLEM

A payment provider receives one currency but must deliver another it does not hold.

Project Nexus requires an FX provider.  
Project Rialto uses a vehicle when direct exchange is unavailable.

*Note:* Project Nexus states that every cross-border, cross-currency payment needs an actor willing and able to exchange the currencies. Project Rialto experimentally uses a vehicle currency when direct exchange is unavailable. The paper studies the intermediary asset observed inside DEX pool routes.

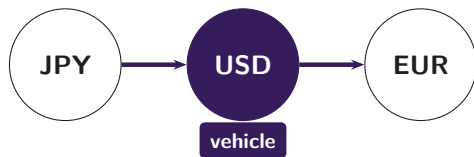


# Pool routes reveal the vehicle currency

## FX ANALOGY

Yen–euro may trade through dollars.

Conventional turnover data show two legs, not the customer's connected route.



Pool logs reveal the path.

**DeFi** is on-chain finance executed by smart contracts; decentralised exchanges (DEXs) trade tokens through liquidity pools. **Stablecoins** are tokens designed to track a reference currency, usually the US dollar.

*Note:* Traditional FX records show turnover in each currency pair, but not whether two legs belong to one customer route. The DEX setting preserves the transaction-level pool path, so the intermediary asset is observed directly.

# The route panel links pool-level swaps

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**472 million**

pool-level swaps

**2,332**

calendar dates

**8**

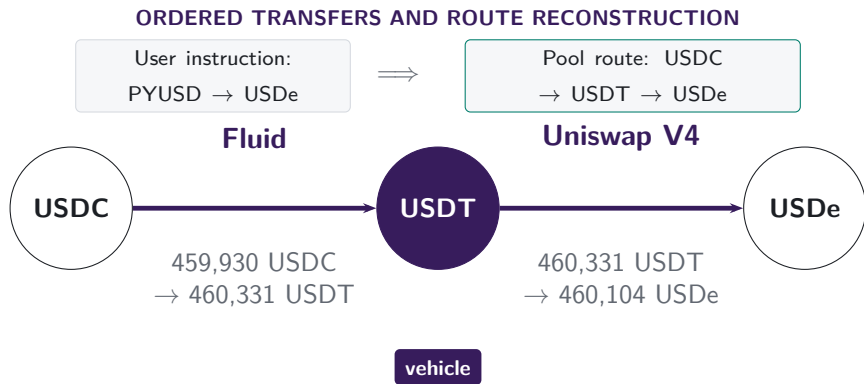
DEX deployments

**February 2020–June 2026**

Uniswap v2, v3, and v4; SushiSwap v2 and v3; Curve; Balancer; and Fluid.

*Note:* The listed deployments form the principal Ethereum route panel. Uniswap v1 enters the separate protocol-architecture analysis because its retained records do not recover endpoint-pair identities.

# Inside the pools, USDT links USDC to USDe



The observed endpoint pair is USDC → USDe. Its legs are USDC → USDT and USDT → USDe; USDT is the vehicle that links them.

*Note:* The route is the connected pool-swap component of transaction 0xbda1d07f...9bf67ad9 on 10 January 2026.

# Vehicle dominance is an intermediary share

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## MEASUREMENT

- Dominance is the share of indirect trade intermediated by asset  $k$ .
- Count dominance weights each route equally; value dominance weights comparable routed value.



### Count-weighted dominance

Share of indirect routes that pass through  $k$

### Value-weighted dominance

Share of comparable routed value that passes through  $k$

# Architecture changes routing opportunities

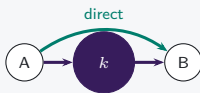
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## V1 mandatory hub



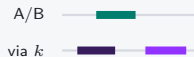
Hub use is imposed

## V2 trading-pair choice



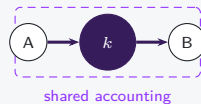
Hub use becomes a  
choice

## V3 range choice



Trading opportunities  
shift

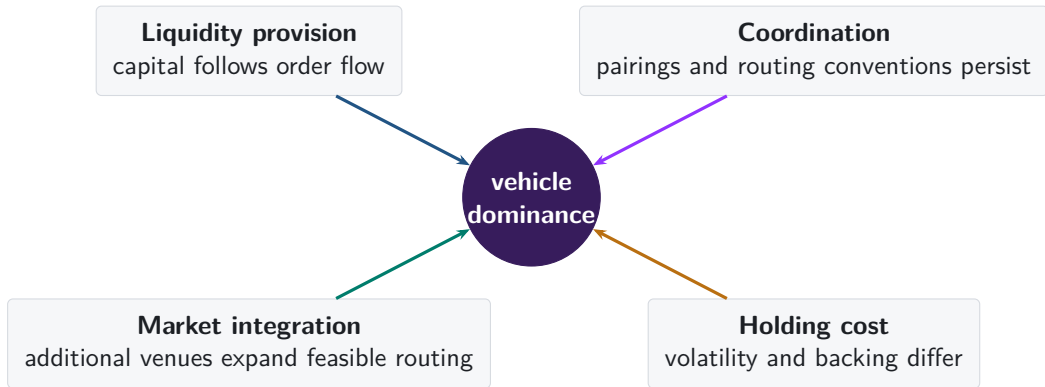
## V4 net settlement



Settlement differs  
from route use

# Four economic forces shape vehicle dominance

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# Native-asset pairing persists after it becomes optional

## V1: PROTOCOL RULE



**217,003 token-to-token trades  
pass through ETH**

Every pool is an ETH-token pool, so the architecture imposes the vehicle.

## V2: MARKET CHOICE

**95.5%**

of single-leg V2 trades use a WETH pool in 2026

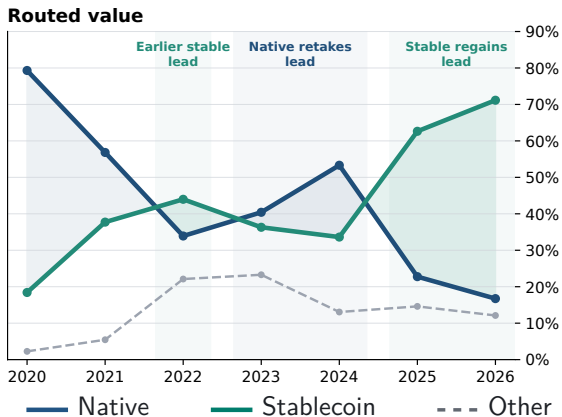
**97.9%**

of endpoint pairs first traded in 2026 include WETH

*Note: V1 links token-to-ETH and ETH-to-token swaps within one transaction. V2 reports single-leg trades and newly traded endpoint pairs on that venue.*

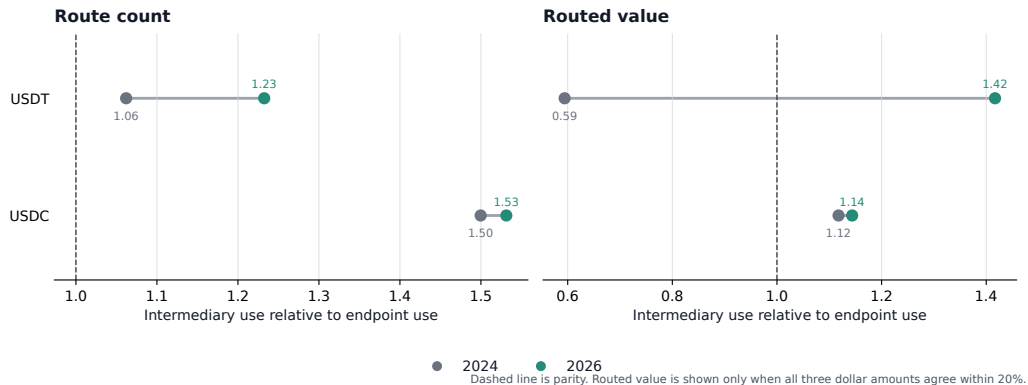
# Stablecoins regain the routed-value lead by 2026

MATCHED JANUARY-JUNE PERIODS



Note: Exact two-leg routes. Routed value requires agreement among source, intermediary, and destination dollar values. The 2026 endpoint covers January through June.

# USDT carries the stablecoin expansion



USDT crosses parity by routed value and extends its count advantage; USDC changes little.

*Note:* Excess use divides intermediary share by route-endpoint share on the same perimeter. The figure compares the same January through June dates in 2024 and 2026.

## Endpoint demand predicts vehicle use within day

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$$I_{a,t} = \alpha + \beta D_{a,t} + \mathbf{A}'_a \theta + \mu_t + \eta_a + \varepsilon_{a,t}$$

$I_{a,t}$  intermediary episode share

$\mu_t$  date effects

$D_{a,t}$  own endpoint-demand share

$\eta_a$  currency effects

$\mathbf{A}_a$  asset-class indicators

*Note:* Outcome and demand are shares of the same day's route universe, measured in percentage points (pp). The estimates are descriptive within-day associations.

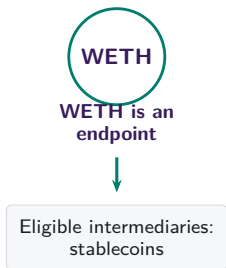
# The demand relationship survives currency and date effects

|                                     | Intermediary share ( $I_{a,t}$ ) [pp] |                   |                    |                      |
|-------------------------------------|---------------------------------------|-------------------|--------------------|----------------------|
|                                     | (1)<br>Pooled                         | (2)<br>Date FE    | (3)<br>+ demand    | (4)<br>+ currency FE |
| Native currency                     | +34.56<br>(23.04)                     | +34.55<br>(23.04) | +0.05<br>(0.56)    | —<br>—               |
| Stablecoin                          | +2.42*<br>(1.32)                      | +2.42*<br>(1.32)  | +0.85*<br>(0.50)   | —<br>—               |
| Endpoint-demand share ( $D_{a,t}$ ) | —<br>—                                | —<br>—            | +1.59***<br>(0.03) | +0.98***<br>(0.30)   |
| Date fixed effects                  | No                                    | Yes               | Yes                | Yes                  |
| Currency fixed effects              | No                                    | No                | No                 | Yes                  |

Endpoint-demand share passes through almost one for one after both fixed effects.

Note: 1,828,862 currency-days and 2,259 dates. Parentheses report currency/date-clustered standard errors. Class coefficients are percentage-point differences; the demand coefficient is percentage points of intermediary share per percentage point of endpoint-demand share. Asterisks \*, \*\*, and \*\*\* denote significance at 10%, 5%, and 1%, respectively.

# WETH-linked endpoint pairs carry the count rotation



**+21.1 pp** All matched endpoint pairs  
WETH-linked route-count mass: 26.6% → 46.8%

**+3.7 pp** Endpoint pairs without WETH endpoints  
79,347 matched groups

Changes within fixed endpoint pairs are +0.2 pp overall and +0.3 pp without WETH endpoints.

*Note:* Exact two-leg native-WETH-versus-stablecoin routes on common month-days. Within-endpoint-pair estimates fix endpoint pair, month-day, and route scope.

# Who trades with whom makes the vehicle

## CONTRIBUTION TO THE STABLECOIN-SHARE INCREASE

**Route count: +25.4 pp**

Other endpoint pairs  +15.2 pp

One native, one stable  +9.2 pp

Two stable endpoints  +1.0 pp

**Routed value: +43.9 pp**

Other endpoint pairs  +20.6 pp

One native, one stable  +10.1 pp

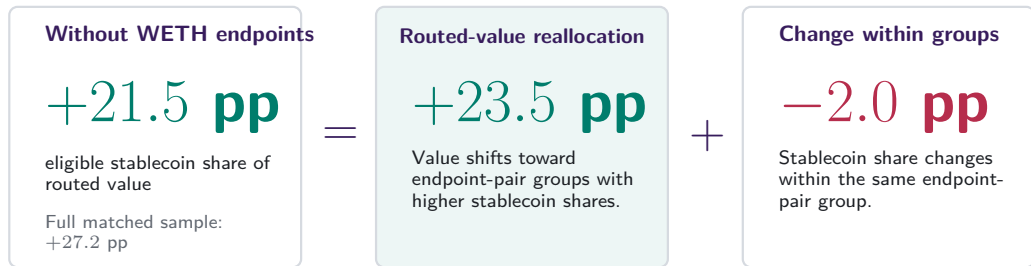
Two stable endpoints  +13.2 pp

USDT supplies +13.7 pp of the +13.2 pp two-stable endpoint value channel.

*Note:* 181 common days. Exact additive decomposition; descriptive.

# Trading shifts toward stablecoin-heavy endpoint pairs

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*Note:* The daily midpoint identity is exact after removing WETH endpoints. Routed value requires agreement among source, intermediary, and destination dollar values.



# New endpoint pairs deliver the largest positive component

EXACT POOLED DECOMPOSITION, 2024 TO 2026

2024 **16.9%** → 2026 **42.5%** **+25.7 pp**  
aggregate rotation

## Continuing endpoint pairs

**−0.1 pp** net within-endpoint-pair change

**+8.6 pp** activity reallocation

## Endpoint-pair entry and exit

**+21.0 pp** newly traded endpoint pairs

**−3.3 pp** endpoint-pair exit

Pair entry contributes +21.0 pp; net switching inside continuing pairs contributes −0.1 pp.

*Note:* Matched January–June periods in 2024 and 2026. The within-pair component combines +1.3 pp toward stablecoins and −1.4 pp toward native assets. The remaining continuing-pair-weight component is −0.5 pp. Components sum using unrounded values.

## Local bridge depth predicts route choice

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**84.1%**

route share captured by  
the deepest bridge

2024 **75.5%**

2026 **86.5%**

deepest-bridge share

**+6.79 pp**

per log point of  
weaker-leg deposited  
capital

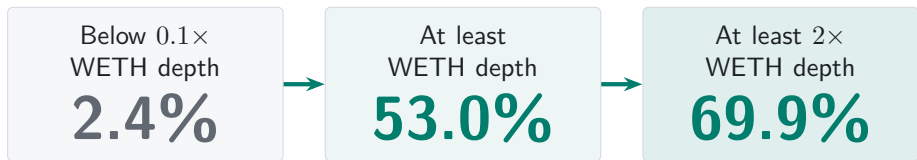
The weaker atomic leg carries the economically relevant depth constraint.

*Note:* The analysis compares five vehicle assets within locally supported endpoint-date-route-scope sets. Depth is  $\log(1 + \text{prior deposited capital in USD})$  on the weaker atomic leg; the regression also controls for vehicle reach.

# Stablecoin penetration rises with relative bridge depth

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## FIRST 30 DAYS AFTER STABLE-BRIDGE AVAILABILITY

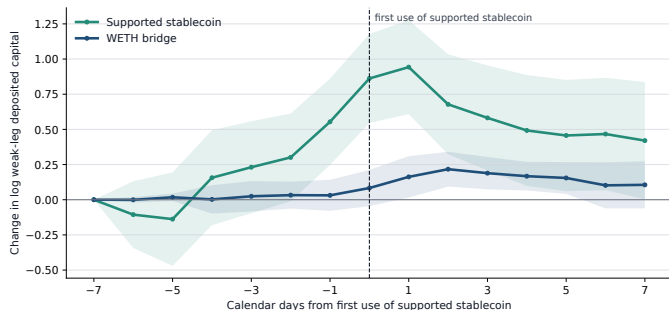


**First-month use of the supported stablecoin: 42.6% below  $0.1\times$  WETH depth, 84.1% at or above it.**

*Note:* The route-share cards use 11,327 endpoint-pair days across 855 events. Adoption gives equal weight to 853 events with positive stable and WETH depth and matches support, depth, and first use to the same stablecoin set. The adjusted gap is +25.2 pp (SE 5.8 pp); with two non-stablecoin endpoints, it is +35.8 pp (SE 11.9 pp). Capital and demand are jointly determined.

# Bridge capital builds before use

## SUPPORTED STABLECOIN FIRST USED AT DAY ZERO



+0.86

stablecoin bridge  
day -7 to 0

+0.78

pre-route change  
relative to WETH

-0.44

stablecoin bridge  
day 0 to +7

Route-specific capital accumulates before use, then partially recedes.

*Note:* Equal-event path for 246 events whose supported stablecoin is first used 7 to 119 days after persistent support. Capital is prior-calendar V2-family deposited capital on the weaker leg; shading gives 95% confidence intervals clustered by ordered endpoint pair and adoption date. The sample conditions on eventual use and does not identify provider intent.

# Most first-use capital sits in pools already active

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SUPPORTED STABLECOIN, DAY -7 TO FIRST USE; MATCHED WETH BENCHMARK

92.5%

of first-use-day capital lies  
in pools active one week  
earlier

+0.16

continuing-pool capital  
growth relative to WETH  
[log points]

+6.9 pp

higher chance of a newly  
active pool relative to  
WETH

Continuing pools hold most capital; pool activation also rises before adoption.

*Note:* Equal-event estimates for 246 first-use events. A newly active pool has positive prior-calendar deposited capital on day zero and none on day -7; continuing pools are positive on both dates. Difference-column inference is two-way clustered by ordered endpoint pair and adoption date, with Holm adjustment across four pool-composition outcomes. Deposited-capital stocks do not identify provider cash flow or intent.

# V4 internal routing forecasts later entry in volatile markets

## 10 PP MORE INTERNAL SAME-ASSET ROUTING

DAYS 1 TO 30

**+0.086\*\*\***

actions by 180-day  
incumbent origins [log  
points]

s.e. 0.024

DAYS 31 TO 120

**+0.153\*\*\***

origins first active after  
measurement [log points]

s.e. 0.032

PERSISTENT VOLATILITY

**+0.318\*\*\***

additional first-active-origin  
response per 1 SD of  
30-day WETH volatility  
[log points]

s.e. 0.039

Incumbents act within a month; first-active origins follow, especially when volatility stays high.

*Note:* 1,107 V4 vehicle-days and 223 date clusters; zero-liquidity updates excluded; 180 prior days required. Volatility is the lagged 30-day mean of realised WETH volatility. Vehicle and date effects; origin-day controls; vehicle and control volatility slopes; date-clustered standard errors. The interaction uses Holm adjustment across two outcomes. A same-vehicle-date V3 comparison does not distinguish the mature-calendar level or volatility slope. Origins proxy for account participation, not LP-position ownership. Predictive associations.

# Vehicle identity survives broader price competition

## SAME INPUT AND PRE-TRANSACTION STATE

Share of routes with more than 1 bp higher gross output

SAME VEHICLE, USED  
VENUES

**6.6%**

**2.0 pp**

additional incidence after opening the  
vehicle set and direct path

SAME VEHICLE, ALL  
EXACT VENUES

**44.4%**

**-1.2 pp**

change in stablecoin vehicle share, s.e.  
0.2 pp

ANY NAMED VEHICLE  
OR DIRECT

**46.4%**

*Note:* 777,651 common-support routes on 73 monthly dates. Exact Uniswap V2, SushiSwap V2, and Uniswap V3 states; coherent values; at least \$100 input; 5% own-price-impact limit. Improvements are gross of gas. The median gain among improved routes is 21.9 bp. Late-sample venue coverage declines as other exchanges expand.

# Dominance forms through entry, depth, and provider response

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- 1 New endpoint pairs carry the rotation and persist.**  
Entry contributes +21.0 pp; 120-day own-stable shares: USDC 88.3%, USDT 87.7%.
- 2 Competitive bridge depth turns support into adoption.**  
30-day adoption: 42.6% below  $0.1 \times$  WETH depth; 84.1% otherwise.
- 3 V4 internal routing: incumbents act first, new origins follow.**  
First-active origins rise  $+0.153^{***}$  in days 31 to 120; persistent volatility adds  $+0.318^{***}$ .

**Dominance forms through market entry, local capital, and provider adjustment.**



# Appendix map

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## Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

## Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

## Robustness

- A7 Timing and venue-scope robustness
- A8 Count and value weighting
- A9 Observed route endpoints

## Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

# Pool data may start after the user instruction

## USER INSTRUCTION RECORDED BY THE EXPLORER

**Transaction details** # 0x

⚡ Swap 460.00K PYUSD for 460.10K USDe on 0x 0x

## THE DECISIVE SETTLEMENT TRANSFERS

USDC ERC-20 Transfer Market Maker: 0x51c7..... → Mainnetsettlr

USDC ERC-20 Transfer Mainnetsettlr → Fluid: Liquidity

The maker supplies the USDC used by the Fluid leg. The instruction is PYUSD → USDe.

*Note:* Transaction 0xbda1d07f...9bf67ad9, 10 January 2026. The data do not reveal whether PYUSD–USDC is another vehicle leg or executor inventory. Disconnected groups comprise 6.2% of reconstructed components in 2026. Treating each internally valid component as a separate route raises stablecoin count dominance from 42.3% to 43.7%.

# A1. One reconstructed route is one unit

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**one route unit**  
two swap legs

- Each linked  $i \rightarrow k \rightarrow o$  route is counted once.
- Asset  $k$  is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by  $k$ .

# How we reconstruct a route from one transaction

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## Direct swap

one transaction



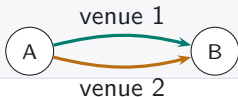
## Sequential vehicle route

one transaction



## Parallel split

one transaction



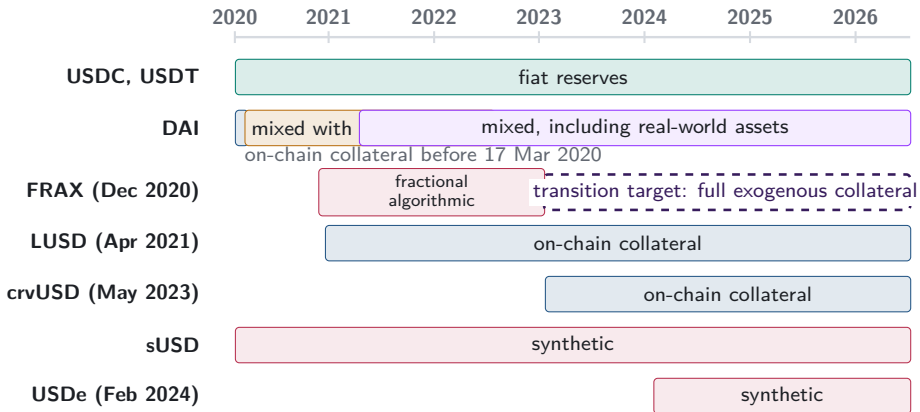
## Round trip

one transaction



Vehicle choice directs indirect order flow toward particular assets and pools. Transaction-level paths make that choice visible inside aggregate turnover.

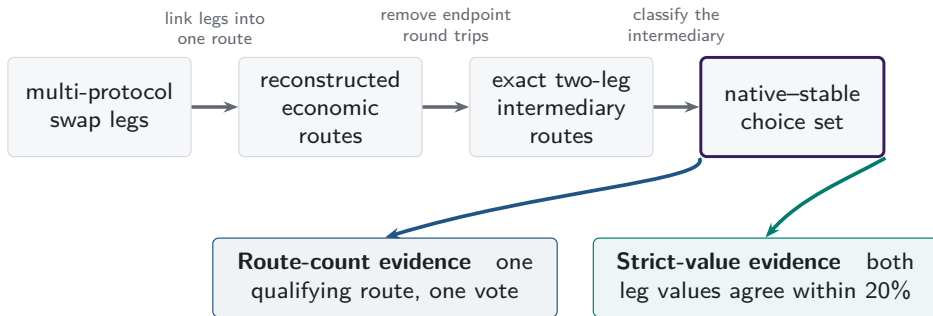
## A2. Stablecoin backing changes over time



*Note:* Bars begin at public launch or the sample start for older tokens. DAI's dated regimes follow the Maker Governance actions that added USDC-A and activated RWA001-A. FRAX's dashed arrow follows FIP-188 and denotes a target, not observed collateral.

## A3. One route universe supports two measurements

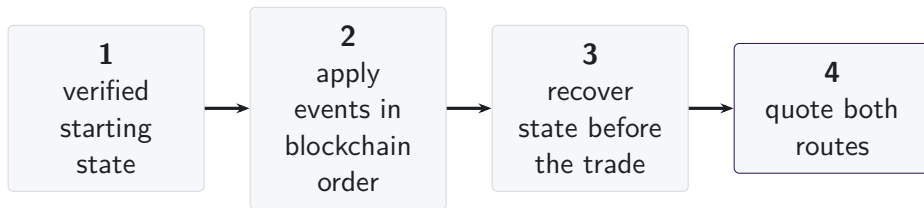
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Token pairs inside multi-asset pools can supply route legs. V1/V2 architecture evidence is separate; dollar support changes value weighting.

## A4. State is reconstructed immediately before execution

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- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

## A5. Each AMM family has a different state vector

|                        | reserves or<br>active liquidity | ticks | amplification | weights or<br>scaling | rates or<br>oracles | fees | hook-dependent<br>pricing |
|------------------------|---------------------------------|-------|---------------|-----------------------|---------------------|------|---------------------------|
| Constant product       | ■                               | —     | —             | —                     | —                   | ■    | —                         |
| Concentrated liquidity | ■                               | ■     | —             | —                     | —                   | ■    | —                         |
| StableSwap families    | ■                               | —     | ■             | —                     | ■                   | ■    | —                         |
| Weighted pools         | ■                               | —     | —             | ■                     | ■                   | ■    | —                         |
| V4 singleton           | ■                               | ■     | —             | —                     | —                   | ■    | □                         |

### Executable-quote validation

■ recovered pricing input

□ hook-dependent pricing unavailable

### Protocol mechanisms studied

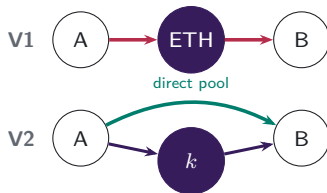
V4 singleton flash accounting and netting remain in scope; A5.3 shows the settlement mechanism.



## A5.1. Pool formation determines available paths

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V1 PROTOCOL RULE  $\rightarrow$  V2 MARKET CHOICE



V1 admits only ETH-token pools. V2 permits any ERC-20 pool, so market creation determines whether direct and vehicle paths coexist.

**Which paths exist?**

**Economic implication: pool creation changes the feasible route set.**

## A5.2. Executable depth comes from active liquidity

### V3 CONCENTRATED-LIQUIDITY POSITIONS



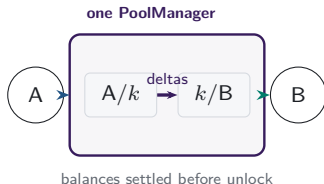
LPs choose a price range and fee tier for each atomic pool.  
Positions covering the current price determine active liquidity  
in the direct pool and each vehicle spoke.

**Which path offers depth at this trade size?**

**Economic implication:** executable depth is path- and size-specific.

## A5.3. Shared accounting moves route settlement

### V4 SINGLETON AND LOCK ACCOUNTING



One PoolManager holds all pools. During a lock, the two route legs update balance deltas; the caller settles its obligations before the lock ends.

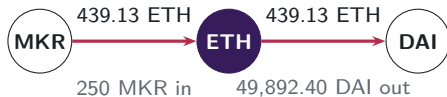
**Where is the route settled?**

**Economic implication: shared accounting changes the settlement boundary.**

## A6. V1 forced routes have an on-chain signature

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- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.
- 129,810 mandate-era token-to-token transactions carry exactly matching ETH legs; the trace shows the largest.

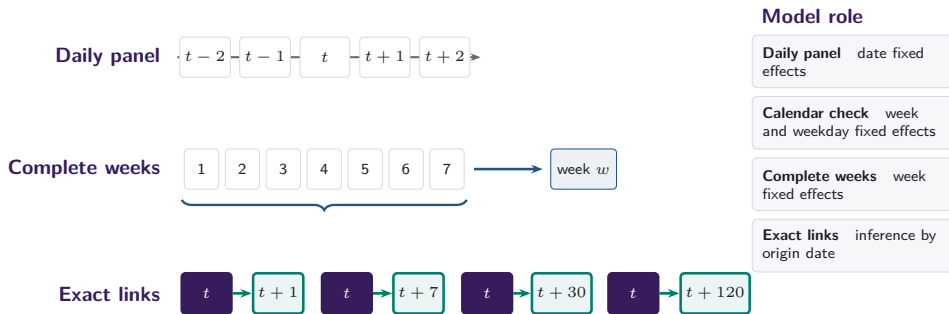


**same transaction, matching ETH flow**

*Note:* Both legs of transaction 0x4dca160d...96a0ca16 (15 March 2020, block 9,674,728) report the same ETH amount to the last recorded digit.

## A7. Daily and weekly frequencies answer different questions

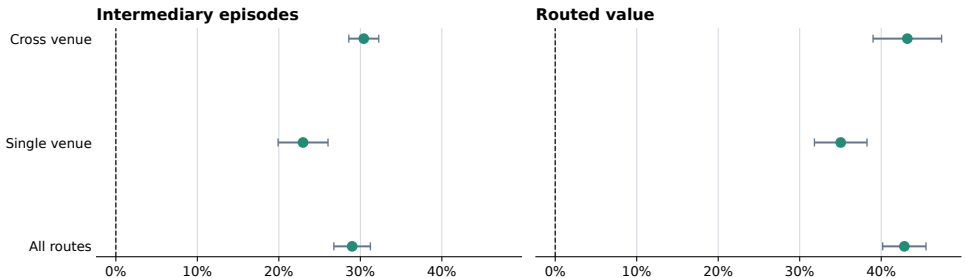
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Exact horizons use the required target calendar date.

## A7b. Stablecoin use rises within every venue scope

### Stable share rises within realised route scopes

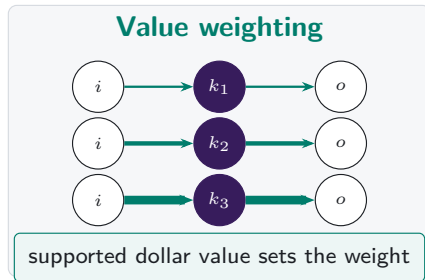
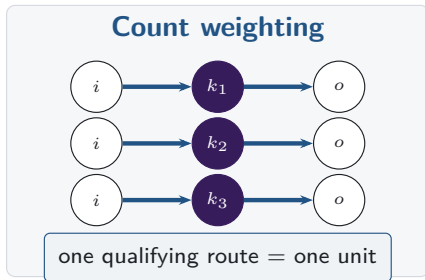


Points are 2024-to-2026 changes; bars are 95% HAC intervals.

*Note:* Changes compare matched January-through-June dates in 2024 and 2026. Bars show calendar-HAC confidence intervals; venue scope records realised routes.

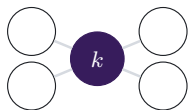
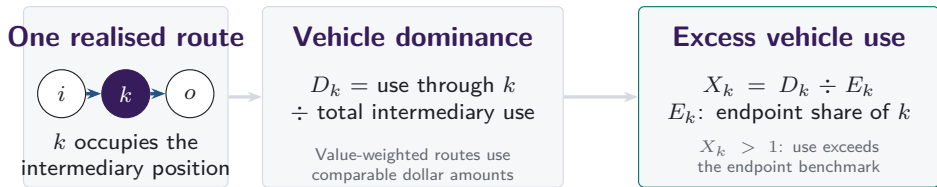
## A8. Count versus value weighting

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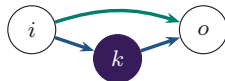


*Note:* Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

# Vehicle dominance aggregates realised intermediary choices



**Network position**



**Execution cost**

Endpoint use is the benchmark; network position and same-state cost remain separate.



## A9. Intermediary and route-endpoint use are separate

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### Intermediary share

$$I_{k,t}^N = \frac{n_{k,t}^I}{\sum_j n_{j,t}^I}$$

How often asset  $k$  sits inside a route.

### Endpoint share

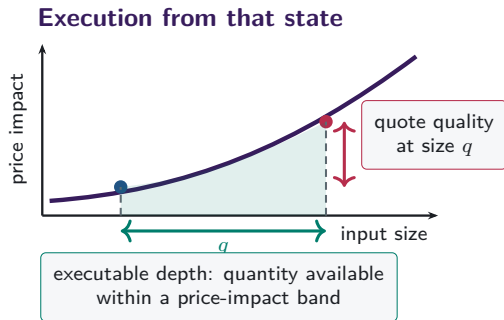
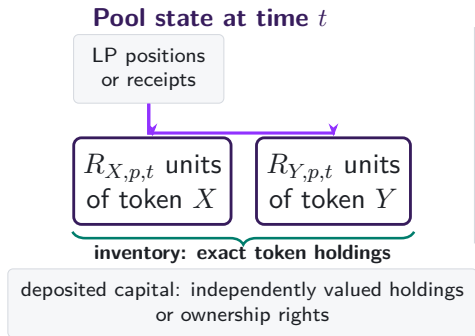
$$E_{k,t}^N = \frac{n_{k,t}^E}{\sum_j n_{j,t}^E}$$

How often asset  $k$  appears at either end of the observed pool route.

Observed pool-route endpoints supply the benchmark.

**Excess use compares intermediation with observed route-endpoint use on the same support.**

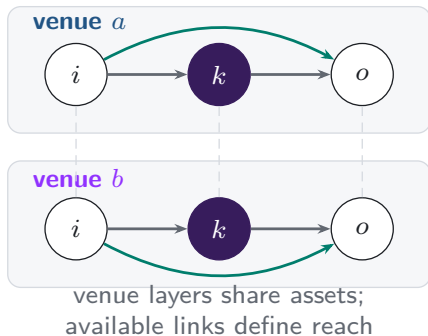
## A10. Capital, inventory, and depth differ



Depth and quotes vary with size and state.

## A11. Additional venues expand the feasible route set

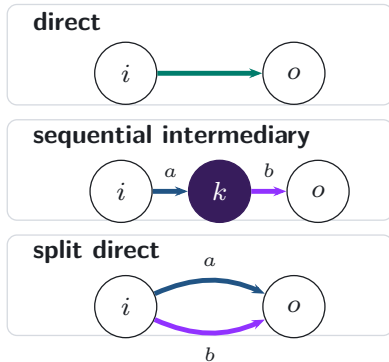
### Feasible venue layers



realised  
execution

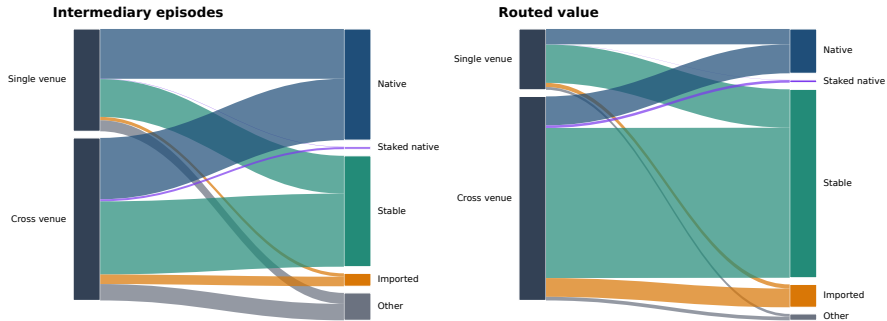


### Realised route topology



## A11b. Venue scope and vehicle type differ in 2026

Integration and vehicle composition are separate margins in 2026



Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

*Note:* Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

## A12. References: vehicle currencies and market structure

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## A13. References: decentralised exchange design

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